PHD125 Pharmaceutical Organic Chemistry · Lipids

Tutorial: exam-style practice

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (38 marks). Same format as the final exam.

Answer Part A by circling one option. Answer Part B in full, showing your working. The answer key is at the back.

Part A · Multiple choice

Choose one answer for each question.

A1 Topic: Lipids CH6-LO1

Fatty acids usually contain:

A. an even number of carbons in an unbranched chain
B. an odd number of carbons
C. a benzene ring
D. a branched chain

A2 Topic: Lipids CH6-LO1

Oleic acid (C18:1) is:

A. monounsaturated
B. saturated
C. polyunsaturated
D. a trans fatty acid

A3 Topic: Lipids CH6-LO1

Arachidonic acid is the starting material for:

A. prostaglandins
B. cholesterol
C. glycogen
D. waxes

A4 Topic: Lipids CH6-LO2

A wax is an ester of:

A. a long-chain fatty acid and a long-chain alcohol
B. glycerol and three fatty acids
C. phosphoric acid and choline
D. cholesterol and glucose

A5 Topic: Lipids CH6-LO2

Most animal fats are solid because they contain mainly:

A. saturated fatty acids
B. cis-unsaturated fatty acids
C. glycerol
D. water

A6 Topic: Lipids CH6-LO3

Partial hydrogenation of vegetable oil can produce:

A. trans fatty acids
B. phospholipids
C. cholesterol
D. glycerol

A7 Topic: Lipids CH6-LO3

Which slows oxidative rancidity?

A. vitamin E (tocopherol)
B. water
C. light
D. copper ions

A8 Topic: Lipids CH6-LO3

Saponification of a fat produces glycerol and:

A. salts of fatty acids (soap)
B. free fatty acids
C. fatty alcohols
D. trans fats

A9 Topic: Lipids CH6-LO4

A molecule with both a hydrophilic and a hydrophobic part is:

A. amphipathic
B. hydrophobic
C. saturated
D. nonsaponifiable

A10 Topic: Lipids CH6-LO4

Sphingomyelin is rich in:

A. the myelin sheath around nerves
B. adipose tissue
C. tooth enamel
D. blood plasma only

A11 Topic: Lipids CH6-LO5

Which is a nonsaponifiable lipid?

A. cholesterol
B. tristearin
C. lecithin
D. carnauba wax

A12 Topic: Lipids CH6-LO5

Bile salts aid digestion by:

A. emulsifying fats into small droplets
B. hydrolysing proteins
C. oxidising glucose
D. forming cholesterol

Part B · Structured questions

Write your answers on paper, then self-mark with the worked solution.

Question B1 Topic: Lipids CH6-LO1 8 marks, C3

Three C18 fatty acids: stearic acid (mp 70 °C), oleic acid (mp 16 °C) and linoleic acid (mp −5 °C).

(a) Classify each as saturated, monounsaturated or polyunsaturated, and give its shorthand notation. (3 marks)

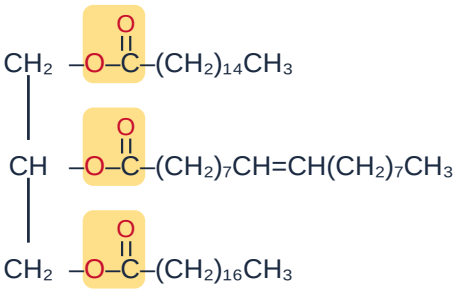
(b) Explain the trend in melting points. (2 marks)

(c) Explain what is meant by an essential fatty acid, and name one ω-3 fatty acid. (2 marks)

(d) State how aspirin is linked to fatty acid chemistry. (1 mark)

Question B2 Topic: Lipids CH6-LO2 8 marks, C4

The structure of a triacylglycerol is shown.



shaded: ester groups

(a) Name the functional group that joins each fatty acid to the backbone, and name the alcohol that forms the backbone. (2 marks)

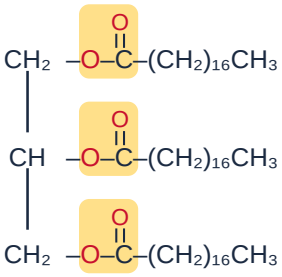
(b) Is this a simple or a mixed triglyceride? Give a reason. (2 marks)

(c) It is completely hydrogenated with H_2/Ni. Which fatty acid chain changes, and what does it become? Explain the effect on the melting point. (2 marks)

(d) Explain how partial hydrogenation can produce trans fats, and why these are a health concern. (2 marks)

Question B3 Topic: Lipids CH6-LO3 8 marks, C3

Tristearin is shown.



shaded: ester groups

(a) Write a balanced equation (condensed formulas or words with formulas) for the saponification of tristearin with sodium hydroxide, and name the products. (3 marks)

(b) Explain how the soap formed removes grease from skin. (3 marks)

(c) Explain why this soap forms a scum in hard water. (1 mark)

(d) Give the products when tristearin is hydrolysed by pancreatic lipase in the gut. (1 mark)

Question B4 Topic: Lipids · CH6-LO4 · 8 marks, C4

(a) Compare the structure of phosphatidylcholine (lecithin) with that of a triacylglycerol. (2 marks)

(b) Explain why phospholipids form a bilayer when placed in water. (2 marks)

(c) Name two components of a cell membrane other than phospholipids, and give the role of each. (2 marks)

(d) Lidocaine is injected as lidocaine hydrochloride, but it must cross nerve membranes as the free base. Explain why the free base crosses more easily. (2 marks)

Question B5 Topic: Lipids · CH6-LO5 · 6 marks, C2

(a) Describe the steroid nucleus. (1 mark)

(b) Give three roles of cholesterol in the body. (3 marks)

(c) Explain why HDL is called "good" cholesterol. (1 mark)

(d) Explain how statins lower blood cholesterol. (1 mark)

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Tutorial: exam-style practice

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Part A: 12 multiple-choice questions (1 mark each). Part B: 5 structured questions (38 marks). Same format as the final exam.

Answer key

Part A · Multiple choice

A1 Topic: Lipids CH6-LO1

Fatty acids usually contain:

A. an even number of carbons in an unbranched chain

B. an odd number of carbons

C. a benzene ring

D. a branched chain

Typically 12–20 carbons (FGOB 23.2).

A2 Topic: Lipids CH6-LO1

Oleic acid (C18:1) is:

A. monounsaturated

B. saturated

C. polyunsaturated

D. a trans fatty acid

One cis C=C.

A3 Topic: Lipids CH6-LO1

Arachidonic acid is the starting material for:

A. prostaglandins

B. cholesterol

C. glycogen

D. waxes

Blocked by NSAIDs via COX.

A4 Topic: Lipids CH6-LO2

A wax is an ester of:

A. a long-chain fatty acid and a long-chain alcohol

B. glycerol and three fatty acids

C. phosphoric acid and choline

D. cholesterol and glucose

FGOB 23.2.

A5 Topic: Lipids CH6-LO2

Most animal fats are solid because they contain mainly:

A. saturated fatty acids

B. cis-unsaturated fatty acids

C. glycerol

D. water

Straight chains pack closely.

A6 Topic: Lipids CH6-LO3

Partial hydrogenation of vegetable oil can produce:

A. trans fatty acids

B. phospholipids

C. cholesterol

D. glycerol

Some cis C=C are isomerised to trans.

A7 Topic: Lipids CH6-LO3

Which slows oxidative rancidity?

A. vitamin E (tocopherol)

B. water

C. light

D. copper ions

An antioxidant.

A8 Topic: Lipids CH6-LO3

Saponification of a fat produces glycerol and:

A. salts of fatty acids (soap)

B. free fatty acids

C. fatty alcohols

D. trans fats

Basic hydrolysis.

A9 Topic: Lipids CH6-LO4

A molecule with both a hydrophilic and a hydrophobic part is:

A. amphipathic

B. hydrophobic

C. saturated

D. nonsaponifiable

Soaps and phospholipids.

A10 Topic: Lipids CH6-LO4

Sphingomyelin is rich in:

A. the myelin sheath around nerves

B. adipose tissue

C. tooth enamel

D. blood plasma only

A sphingolipid.

A11 Topic: Lipids CH6-LO5

Which is a nonsaponifiable lipid?

A. cholesterol

B. tristearin

C. lecithin

D. carnauba wax

Steroids have no ester bonds.

A12 Topic: Lipids CH6-LO5

Bile salts aid digestion by:

A. emulsifying fats into small droplets

B. hydrolysing proteins

C. oxidising glucose

D. forming cholesterol

They are amphipathic (FGOB 23.6).

Part B · Structured questions

Question B1 Topic: Lipids CH6-LO1 8 marks, C3

Three C18 fatty acids: stearic acid (mp 70 °C), oleic acid (mp 16 °C) and linoleic acid (mp −5 °C).

(a) Classify each as saturated, monounsaturated or polyunsaturated, and give its shorthand notation. (3 marks)

(b) Explain the trend in melting points. (2 marks)

(c) Explain what is meant by an essential fatty acid, and name one ω-3 fatty acid. (2 marks)

(d) State how aspirin is linked to fatty acid chemistry. (1 mark)

(a) Stearic acid: saturated, C18:0. Oleic acid: monounsaturated, C18:1. Linoleic acid: polyunsaturated, C18:2.

(b) Each cis C=C puts a kink in the chain, so the chains cannot pack closely; dispersion forces are weaker and the melting point falls as the number of C=C increases.

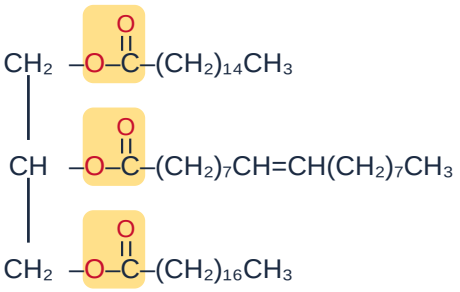
(c) A fatty acid the body cannot make, so it must come from the diet. ω-3: α-linolenic acid (or EPA or DHA).

(d) Aspirin blocks cyclooxygenase, the enzyme that converts arachidonic acid (a fatty acid) into prostaglandins.

Step	Marking criteria	Marks
1	Stearic: saturated, C18:0	1
2	Oleic: monounsaturated, C18:1	1
3	Linoleic: polyunsaturated, C18:2	1
4	Cis C=C kinks the chain; poor packing	1
5	Weaker dispersion forces → lower mp	1
6	Essential: cannot be made by the body	1
7	Correct ω-3 example	1
8	Aspirin blocks COX / prostaglandin synthesis	1
	Total	**8**

Question B2 Topic: Lipids CH6-LO2 8 marks, C4

The structure of a triacylglycerol is shown.



shaded: ester groups

(a) Name the functional group that joins each fatty acid to the backbone, and name the alcohol that forms the backbone. (2 marks)

(b) Is this a simple or a mixed triglyceride? Give a reason. (2 marks)

(c) It is completely hydrogenated with H_2/Ni. Which fatty acid chain changes, and what does it become? Explain the effect on the melting point. (2 marks)

(d) Explain how partial hydrogenation can produce trans fats, and why these are a health concern. (2 marks)

(a) Ester groups; glycerol.

(b) Mixed: it contains three different fatty acids (palmitic, oleic and stearic).

(c) The oleic acid chain (with the C=C) gains H_2 and becomes a stearic acid chain; the product has only saturated chains, which pack closely, so its melting point rises.

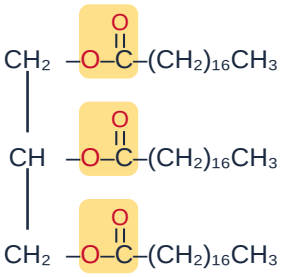
(d) During partial hydrogenation some cis C=C bonds that are not hydrogenated are converted to the trans form; trans fats raise LDL cholesterol and the risk of heart disease.

Step	Marking criteria	Marks
1	Ester groups	1
2	Glycerol	1
3	Mixed	1
4	Reason: three different fatty acids	1
5	Oleic chain → stearic chain	1
6	Higher melting point (saturated, packs well)	1

7	Cis → trans during partial hydrogenation	1
8	Raise LDL / heart disease risk	1
	Total	**8**

Question B3 Topic: Lipids CH6-LO3 8 marks, C3

Tristearin is shown.



shaded: ester groups

(a) Write a balanced equation (condensed formulas or words with formulas) for the saponification of tristearin with sodium hydroxide, and name the products. (3 marks)

(b) Explain how the soap formed removes grease from skin. (3 marks)

(c) Explain why this soap forms a scum in hard water. (1 mark)

(d) Give the products when tristearin is hydrolysed by pancreatic lipase in the gut. (1 mark)

(a) Tristearin + 3 NaOH → glycerol + 3 $CH_3(CH_2)_{16}COO^-Na^+$; the products are glycerol and sodium stearate (soap).

(b) The stearate anion is amphipathic: a long nonpolar tail and an ionic $-COO^-$ head. The tails dissolve in the grease, the heads face the water, forming micelles; the charged micelles repel each other and are rinsed away.

(c) Ca^{2+} and Mg^{2+} ions in hard water form insoluble calcium and magnesium stearate.

(d) Glycerol and stearic acid.

Step	Marking criteria	Marks
1	Reactants: tristearin + 3 NaOH	1
2	Products: glycerol + 3 sodium stearate	1
3	Products named	1
4	Amphipathic: nonpolar tail, ionic head	1
5	Tails in grease, heads in water: micelle	1
6	Micelles rinsed away (repel each other)	1
7	Ca^{2+}/Mg^{2+} form insoluble salts	1

8	Glycerol + stearic acid	1
	Total	**8**

Question B4 Topic: Lipids CH6-LO4 8 marks, C4

(a) Compare the structure of phosphatidylcholine (lecithin) with that of a triacylglycerol. (2 marks)

(b) Explain why phospholipids form a bilayer when placed in water. (2 marks)

(c) Name two components of a cell membrane other than phospholipids, and give the role of each. (2 marks)

(d) Lidocaine is injected as lidocaine hydrochloride, but it must cross nerve membranes as the free base. Explain why the free base crosses more easily. (2 marks)

(a) Both have a glycerol backbone with ester-linked fatty acids. A triacylglycerol has three fatty acids; phosphatidylcholine has two, and its third –OH carries a phosphate joined to choline, a charged polar head.

(b) Phospholipids are amphipathic: the polar heads face the water on both sides, while the nonpolar tails turn inwards away from water, forming a double layer.

(c) Any two: cholesterol (keeps membrane fluidity in range); integral proteins (channels, carriers, receptors); peripheral proteins (signalling, support); sugar chains (cell recognition).

(d) The free base is uncharged, so it can dissolve in and pass through the hydrophobic core of the bilayer; the protonated (charged) form cannot.

Step	Marking criteria	Marks
1	Triglyceride: 3 fatty acids	1
2	Phospholipid: 2 fatty acids + phosphate–choline head	1
3	Amphipathic: heads to water	1
4	Tails inside, away from water → bilayer	1
5	First component and role	1
6	Second component and role	1
7	Free base is uncharged	1
8	Can pass through hydrophobic core; ion cannot	1
	Total	**8**

Question B5 Topic: Lipids CH6-LO5 6 marks, C2

(a) Describe the steroid nucleus. (1 mark)

(b) Give three roles of cholesterol in the body. (3 marks)

(c) Explain why HDL is called "good" cholesterol. (1 mark)

(d) Explain how statins lower blood cholesterol. (1 mark)

(a) Four fused rings: three six-membered (A, B, C) and one five-membered (D).

(b) Any three: component of cell membranes; starting material for bile acids/salts; for steroid hormones (cortisol, aldosterone, sex hormones); for vitamin D.

(c) HDL carries excess cholesterol from tissues back to the liver for removal, lowering heart-disease risk.

(d) They inhibit HMG-CoA reductase, an enzyme the liver needs to make cholesterol.

Step	Marking criteria	Marks
1	Four fused rings (3 six-, 1 five-membered)	1
2	Role 1	1
3	Role 2	1
4	Role 3	1
5	HDL carries cholesterol to the liver	1
6	Statins inhibit HMG-CoA reductase	1
	Total	**6**

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